



Update Cylinder Lubrication Guidelines for Alpha Lubricator

ENGINE TYPE: All

1. Introduction

As for cylinder lubrication guidelines for Alpha lubricator, the service data, HSD-B13102 has been already issued. In order to correspond more specifically to the engine designs and the fuel oil sulphur in use, the previous guidelines including the applicable cylinder oil are overall updated to focus on countermeasures against the cold corrosion wear. The summary of the revision is as follows. We recommend you to refer the following pages about the details and follow these guidelines instead of the previous HSD-B13102 after this.

- 1) The guideline for the current design engines (Mk 8.1 and newer), which could have more possibilities of risk of cold corrosion, is differentiated from that for the previous design engines (Mk 8 and older).
- 2) As the low load operation has become usual, various special tunings such as T/C cut-out, EGB (Exhaust Gas By-pass), WHR (Waste Heat Recovery) and VT (Variable Turbine) are applied for purpose of the fuel consumption improvement during the low load operation. In this condition, there are more possibilities of cold corrosion wear even on the previous design engines as well as the current design engines. Therefore, the guideline of the previous design engines is differentiated depending on whether such special tuning is applied.
- 3) As the supply system of BN100 is getting ready and from an economic point of view, the standard of the applicable cylinder oil is specified as BN40, BN70 and BN100 depending on the engine designs and the fuel oil in use. In Addition, the treatment for BN70-85 is also described as the tentative one until the supply system of BN100 is established.
- 4) As per so-called middle range, BN50-60, where treatment had not been defined yet, the application condition is now defined and the application range is limited.

REMARKS :

2. Selection of the cylinder oil

For each category, the applicable cylinder oil is shown in the table 1.

The worldwide brands for cylinder oil are shown in the table 5.

Table 1 General guidelines to select the cylinder oil

Fuel oil	Sulphur content(S) %	Mk 8 and older engine		Mk 8.1 and newer engine	
Application of special tuning ¹⁾		No	Yes	No	Yes
Viscosity grade		SAE50			
Low sulphur fuel oil	S<0.1	BN40			
	0.1≤S<1.5	BN40			
		BN50-60			
		BN70 ²⁾			
		BN80-85 ²⁾			
		BN100 ²⁾			
High sulphur fuel oil	1.5≤S≤3.5	BN50-60 ³⁾	—	—	
		BN70	BN70-85 ⁵⁾		
		BN80-85 ⁴⁾			
		BN100 ⁶⁾	BN100		

- 1) T/C cut-out, EGB (Exhaust Gas By-pass), WHR (Waste Heat Recovery) and VT (Variable Turbine)
- 2) Limited use for 1-2 week only
- 3) Usable on Mk 7 and older engine only
Please beware that its use is not approved for Mk 8 and newer engine.
- 4) Decrease of the feed rate in proportional with 70 / (80 or 85)
- 5) Increase of the feed rate in proportional with 100 / (70-85) until the supply system of BN100 becomes established.
- 6) Decrease of the feed rate in proportional with 70/100

3. Guidelines for the cylinder oil feed rate

Guidelines for each category are as shown in Table 2.

Table 2 Guidelines for the cylinder oil feed rate

	Mk 8 and older engine	Mk 8.1 and newer engine
Applicable cylinder oil	Select referring Table 1	
ACC setting	034-0.20 (in case of BN70)	0.40-0.20 (in case of BN100)
Minimum feed rate	0.6 g/kWh	
Maximum feed rate during running-in	1.7 g/kWh	
Part-load control	At more than 25% load, control is ordinarily proportional with load. At less than 25% load, control is changed to proportional with rpm. Break point is subject to change based on actual service experience or engine tuning such as WHR, EGB and low load operation.	
Running-in new or reconditioned liners and new piston rings	Feed rate:	
	Cylinder oil in use: BN70	Cylinder oil in use: BN100
	1) Initial feed rate is 1.7 g/kWh. Keep this feed rate for the following hours depending on the piston ring specification. Alu-coat piston ring : 5 hours Non-coat/Hard-coat piston ring : 16 hours	
	2) After the above defined hours, until 500 hours With checking cylinder condition, reduce the feed rate in a stepwise manner, from 1.5 to 0.6 g/kWh (0.2 g/kWh per 100 hours) or ACC factor × fuel oil S%, whichever the higher feed rate. M/E load: Follow the schedule of Fig. 3.	
Familiarizing ACC factor	Starting at 0.34 g/kWh	Starting at 0.40 g/kWh
	Reduce by 0.04g/kWh × S% for minimum 600 hours, with checking the cylinder condition carefully. The minimum feed rate should be strictly complied at 0.6 g/kWh.	
	ACC setting for BN70 cylinder oil in use: Refer Table 3	ACC setting for BN100 cylinder oil in use: Refer Table 4
	In case of the cylinder oil other than the above defined one, adjust cyl. oil feed rate by multiplying ACC factor with "70/BN in use" as shown in the note of this table. As per the low sulphur fuel oil containing the sulphur less than 1.5%, there is a case that ACC can't be actually applied, because the feed rate at ACC familiarisation start is less than the minimum feed rate, 0.6 g/kWh. The cylinder oil feed rate reduction schedule including the fixed feed rate to ACC process is as shown in Fig. 1 and Fig. 2.	
Running-in new rings in already run-in and well running liners (Standard cylinder oil in use)	Feed rate: Increase the feed rate 0.9 g/kWh for 24 hours operation. If the fuel oil sulphur and applied ACC factor combination results in a specific feed rate are higher than 0.9 g/kWh, no extra lubrication is needed. M/E load:	
	Alu-coat	No limit
	Non-coat/Hard-coat	Follow the schedule in Fig. 3
Manoeuvring and load change situations	During starting, manoeuvring and load changes, increase feed rate by means of the "LCD" by 25% of the actual figure and kept at this level for 30 minutes after the load has stabilised.	
Lubrication of cylinders that show abnormal conditions.	Frequent scavenge port inspections of piston rings and cylinder liners are very important for maintaining a safe cylinder condition. If irregularities are observed, adjustments of the lube oil feed rate should be considered. In case of scuffing or sticking piston rings, raise the feed rate to 1.20 g/kWh and lower the Pmax and mep. As soon as the situation has been stabilised, set the lubrication feed rate, Pmax and mep back to normal. In case of high corrosive wear, the ACC factor is to be increased to the highest ACC factor (0.34 for BN70, 0.40 for BN100) and be reduced in a stepwise manner of only 0.02 g/kWh×S% when the wear has been confirmed as normal.	

Table 3 ACC setting for BN70

ACC factor					feed rate (ACC factor× S%)*	HMI setting
0.20	0.22	0.26	0.30	0.34		
Sulphur content % of the fuel oil in use					g/kWh	
0.0	0.0	0.0	0.0	0.0	0.60**	55
0.9	0.8	0.7	0.6	0.5	0.60**	55
1.7	1.5	1.3	1.1	1.0	0.60**	55
1.9	1.7	1.4	1.2	1.1	0.60**	55
2.1	1.9	1.6	1.4	1.2	0.60**	55
2.2	2.0	1.7	1.5	1.3	0.60**	55
2.4	2.2	1.8	1.6	1.4	0.60**	55
2.6	2.3	2.0	1.7	1.5	0.60**	55
2.7	2.5	2.1	1.8	1.6	0.60**	55
2.9	2.6	2.2	1.9	1.7	0.60**	55
3.1	2.7	2.3	2.0	1.8	0.61	55
3.3	3.0	2.5	2.2	1.9	0.65	59
3.4	3.1	2.6	2.3	2.0	0.68	62
3.5	3.2	2.7	2.4	2.1	0.71	65
	3.4	2.9	2.5	2.2	0.75	68
	3.5	3.0	2.6	2.3	0.78	71
		3.2	2.7	2.4	0.82	75
		3.3	2.8	2.5	0.85	77
		3.4	2.9	2.6	0.88	80
		3.5	3.1	2.7	0.92	84
			3.2	2.8	0.95	86
			3.3	2.9	0.99	90
			3.4	3.0	1.02	93
			3.5	3.1	1.05	95
				3.2	1.09	99
				3.3	1.12	102
				3.4	1.16	105
				3.5	1.19	108

- * In case the cylinder oil other than BN70 is used:
For example, when the applied ACC factor is 0.34, the sulphur content is 1.2 % and the cylinder oil in use is BN70, the cylinder oil feed rate is $0.34 \times 1.2 = 0.41$ g/kWh. However, the minimum feed rate is limited at 0.6 g/kWh as noted in **, then the feed rate becomes 0.6 g/kWh.
In case the cylinder oil BN40 is used in this condition, $0.34 \times 1.2 \times 70/40 = 0.71$ g/kWh.
- ** In order to form proper oil film, the feed rate should not be lower than the minimum feed rate, 0.6 g/kWh at any case.

Table 4 ACC setting for BN100

ACC factor						feed rate (ACC factor× S%)*	HMI setting
0.20	0.24	0.28	0.32	0.36	0.40		
Sulphur content % of the fuel oil in use						g/kWh	
0.0	0.0	0.0	0.0	0.0	0.0	0.60**	55
1.0	0.8	0.7	0.6	0.6	0.5	0.60**	55
2.0	1.7	1.4	1.3	1.1	1.0	0.60**	55
2.2	1.8	1.6	1.4	1.2	1.1	0.60**	55
2.4	2.0	1.7	1.5	1.3	1.2	0.60**	55
2.6	2.2	1.9	1.6	1.4	1.3	0.60**	55
2.8	2.3	2.0	1.8	1.6	1.4	0.60**	55
3.0	2.5	2.1	1.9	1.7	1.5	0.60**	55
3.2	2.7	2.3	2.0	1.8	1.6	0.64	58
3.4	2.8	2.4	2.1	1.9	1.7	0.68	62
3.5	3.0	2.6	2.3	2.0	1.8	0.72	65
	3.2	2.7	2.4	2.1	1.9	0.76	69
	3.3	2.9	2.5	2.2	2.0	0.80	73
	3.5	3.0	2.6	2.3	2.1	0.84	76
		3.1	2.8	2.4	2.2	0.88	80
		3.3	2.9	2.6	2.3	0.92	84
		3.4	3.0	2.7	2.4	0.96	87
		3.5	3.1	2.8	2.5	1.00	91
			3.3	2.9	2.6	1.04	95
			3.4	3.0	2.7	1.08	98
			3.5	3.1	2.8	1.12	102
				3.2	2.9	1.16	105
				3.3	3.0	1.20	109
				3.4	3.1	1.24	113
				3.5	3.2	1.28	116
					3.3	1.32	120
					3.4	1.36	124
					3.5	1.40	127

- * In case the cylinder oil other than BN100 is used:
For example, when the applied ACC factor is 0.40, the sulphur content is 2.4 % and the cylinder oil in use is BN100, the cylinder oil feed rate is $0.40 \times 2.4 = 0.96$ g/kWh.
In case the cylinder oil BN80 is used in this condition, $0.40 \times 2.4 \times 100/80 = 1.20$ g/kWh.
- ** In order to form proper oil film, the feed rate should not be lower than the minimum feed rate, 0.6 g/kWh at any case.

Table 5 Recommended worldwide brands for cylinder oil

Oil company	Brand	BN level
Aegean	Alfacylo 540 LS	40
	Alfacylo 570	70
	Alfacylo 100 HS	100
BP	CL-DX 405	40
	CL 505	50
	CL 605	60
	CLO-50M	70
	Energol CL 805	80
	Energol CL 100ACC	100
Castrol	Cyltech 40SX	40
	Cyltech 50S	50
	Cyltech 70	70
	Cyltech 80 AW	80
	Cyltech CL 100ACC	100
Chevron	Taro Special HT LS 40	40
	Taro Special HT 55	55
	Taro Special HT 70	70
	Taro Special HT 100	100
ExxonMobil	Mobilgard L540	40
	Mobilgard 560 VS	60
	Mobilgard 570	70
	Mobilgard 5100	100
Gulf Oil Marine	GulfSea Cylcare DCA 5040H	40
	GulfSea Cylcare EHP 5070	70
	GulfSea Cylcare 8550	85
	GulfSea Cylcare 50100	100
JX Nippon Oil & Energy	Marine C405	40
	Marine C705	70
	Marine C1005	100
Lukoil	Navigo 40 MCL	40
	Navigo 70 MCL	70
	Navigo 100 MCL	100
Shell	Alexia S4	60
	Alexia 50	70
	Alexia S5	80
	Alexia S6	100
Sinopec	Marine Cylinder Oil 5040	40
	Marine Cylinder Oil 5070E	70
Total	Talusia LS 40	40
	Talusia Universal	57
	Talusia HR 70	70
	Talusia Universal 100	100
Idemitsu	Daphne Seamaster A50N	70
Cosmo	Cosmo Marine 5070	70

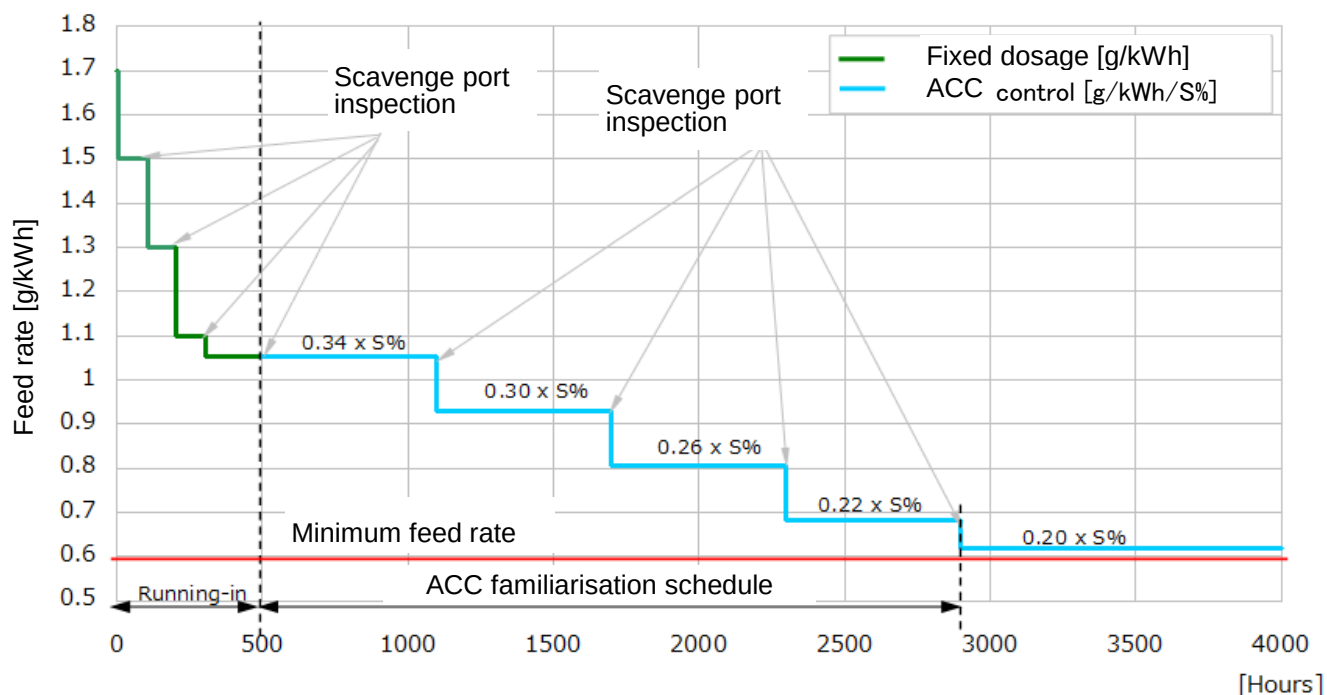


Fig. 1 Reducing feed rate schedule for Mk 8 and older engine
(in case of Fuel oil sulphur: 3.1% Cylinder oil BN 70)

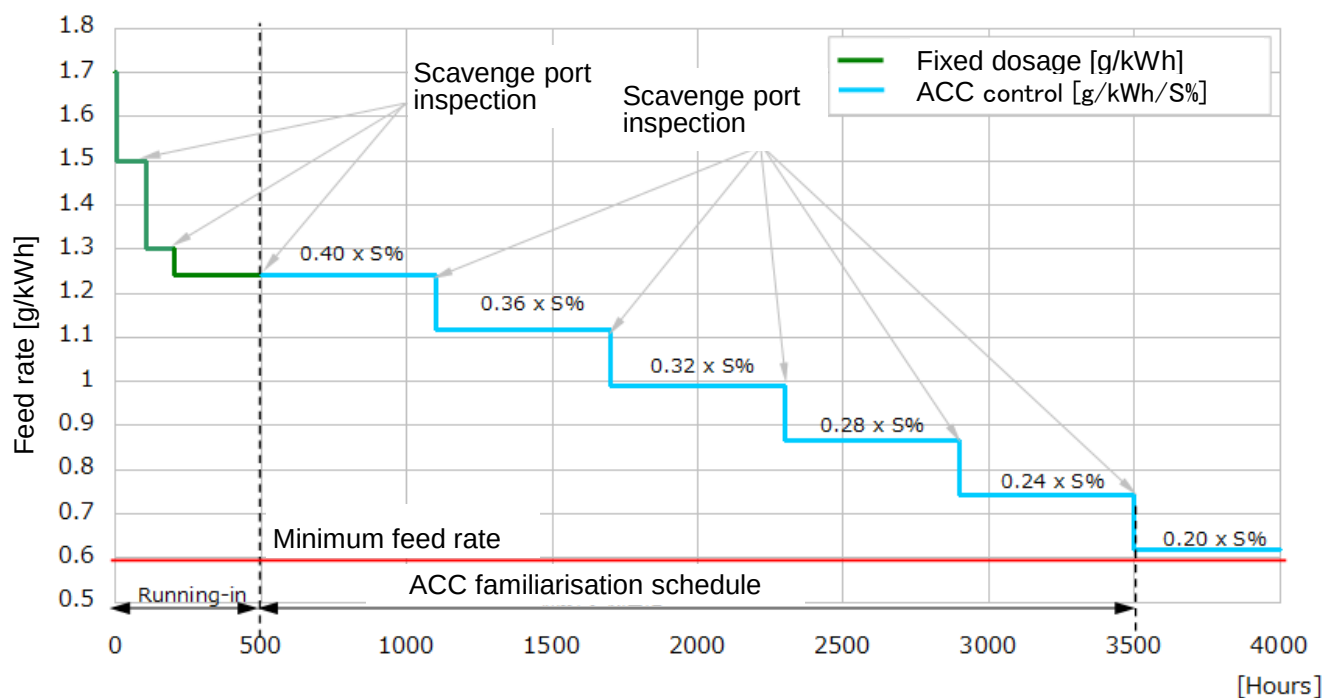


Fig. 2 Reducing feed rate schedule for Mk 8.1 and newer engine
(In case of Fuel oil sulphur: 3.1% Cylinder oil BN 100)

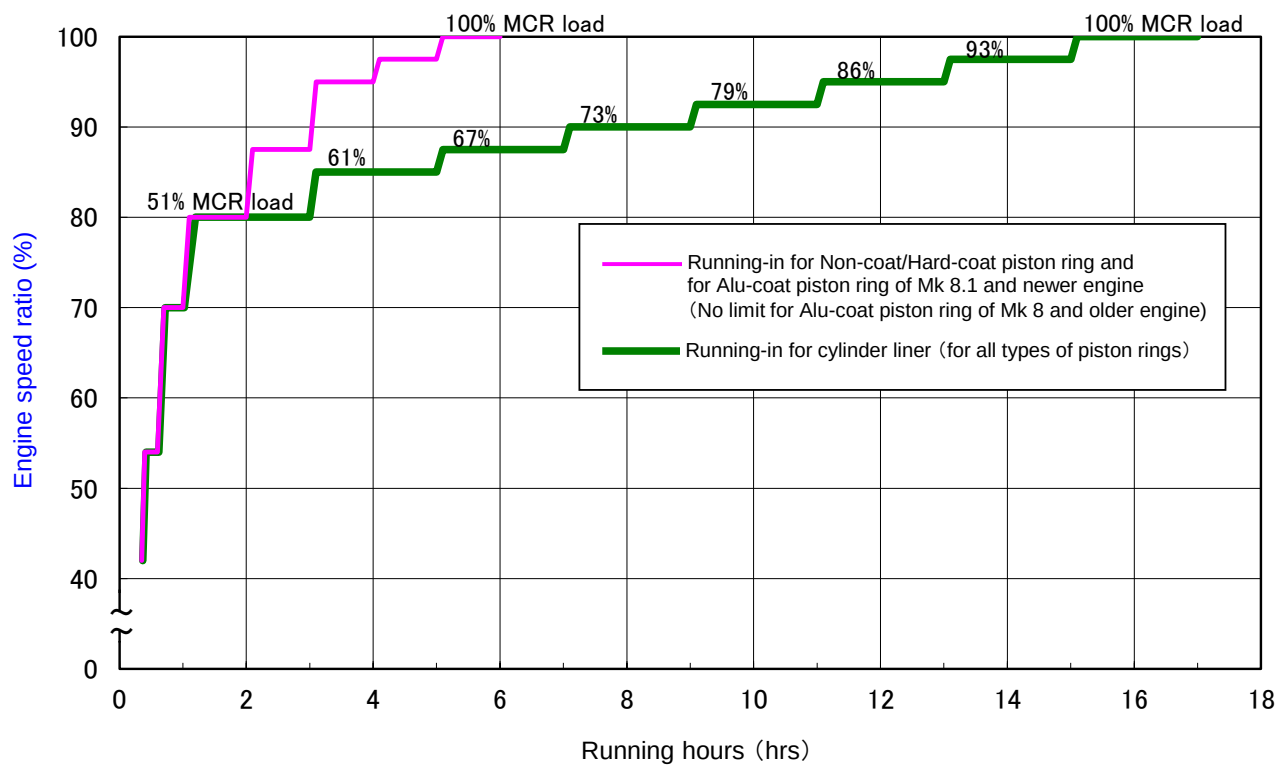


Fig. 3 Recommended load-up schedule at running-in